

# A Drilling and Safety Toolkit for the Rig Floor

Product story · feature highlights · engineering decisions

⚠ FOR UNDERSTANDING PURPOSES ONLY

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Flutter / Dart

Flutter · Dart

Offline-first

PDF & Excel output

Dark rig-friendly design

## 1. At a Glance

DrillTick is a mobile and desktop toolkit that puts the calculators, simulators, forms and reference guides of a drilling site into one app. A kill sheet, a work permit, a safety checklist or a crane lift plan can be filled in on a phone, saved, and exported as a clean PDF, all without an internet connection.

2

main areas

4

well-control tools

7

checklists

6

work permits

181k

lines of Dart

|               |                                                                                       |
|---------------|---------------------------------------------------------------------------------------|
| Project type  | Personal learning project, designed, built and documented end to end by one developer |
| Built with    | Flutter and Dart                                                                      |
| Data model    | Offline-first: a local database on the device, with no accounts and no servers        |
| Output        | A4 PDFs for sheets, permits, checklists and reports; Excel and CSV exports            |
| Scale         | 46 source files and about 181,000 lines of code                                       |
| Documentation | 27 guides, one for each part of the app                                               |

## 2. The Problem

Drilling work runs on paper: kill sheets, permits, inspection checklists, incident forms and lifting plans. Each has its own layout, its own calculations and its own signatures. In the field this causes real friction.

- **Calculations done by hand** under pressure, where one rounding mistake can matter.
- **Forms that are slow to fill in** and hard to read once photocopied.
- **Records scattered** across folders, with no single view of how safety is trending.

- **Training is hard to rehearse.** Surveys, downlinks and sensor calibration are normally practised on live equipment.

### 3. The Solution

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One app with two areas, **Operations** and **Safety**, built around a consistent screen pattern, so every tool feels familiar once the first one has been used.

| Theme              | What it gives the user                                                                                                                                                 |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Well control       | IWCF-style kill sheets for surface and subsea wells, with live calculations and a PDF that closely matches the paper form; slow-rate pressure (SCR) recording for both |
| MWD training tools | Hands-on simulations of taking a survey, downlinking to a tool and wiring and calibrating the rig sensors                                                              |
| Safety management  | Event reporting with a risk meter, a seven-page performance dashboard, a monthly HSE report, seven checklists and six permit-to-work forms                             |
| Lifting operations | A crane lifting plan with load chart, daily check and tandem lifts; a visual lifting manual; and a rigging register with inspections and sling calculations            |

### 4. Feature Highlights

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#### Calculate as you type

- The Surface Kill Sheet runs **eight calculations** live, in dependency order, and the Subsea version runs **nine**, adding choke-line friction and the initial dynamic casing pressure.
- Calculated cells are visibly read-only, so a result can't be overwritten by accident.
- Wherever a number sets a limit, it is rounded in the safe direction: strokes and kill mud weight round **up**, maximum allowable mud weight and MAASP round **down**.
- Two drawings are made by the app itself: a well diagram that follows the entered depths, and a step-down pressure chart.

#### Rehearse before the real thing

- A survey simulation with a guided banner: release the torque, stop the pump, wait, restart, then read the azimuth, inclination and magnetic dip.
- A downlinking simulator with three methods: RPM modulation, mud-flow modulation and wired-pipe commands.
- A rig-sensor calibration module covering hook load, drawworks, pump strokes and standpipe pressure.

## Paperwork that behaves

- Six permits to work share one five-section form: work description, hazards and controls, gas testing, hazard-analysis checklist, and authorisation with open, renew and close.
- Seven checklists with single-tap statuses, custom rows and PDF export.
- A **composite permit** export bundles saved permits, lift plans and tandem plans from different places into one PDF.

## See the whole safety picture

- Incident, near-miss and observation forms with an automatic **risk meter** and an automatic **High Potential Incident** check.
- A dashboard with seven pages, an Excel-style event register with cascading filters, and drill-through from any chart.
- A monthly HSE report with automatic man-hours, frequency rate and severity rate.

## Lift with confidence

- A crane plan that reads the capacity from a load chart, works out the percentage of capacity and ground pressure, and reports Safe, Caution, Critical or **No Lift**.
- A rigging register that tracks inspections and next-due dates, with checklists for wire rope, web, chain and shackles.
- A sling calculator covering six hitch types, with colour-coded angles.

## 5. A Day with the App

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1. **Rig-up.** Run the Pre-Spud checklist for the site and the Drops checklist on the derrick. Brief a new crew member with the Safety Orientation checklist.
2. **Before the job.** Raise a hot work permit for welding, record the gas-test readings and export the PDF.
3. **Lifting.** Complete the crane's daily check, prepare the lift plan against the load chart, and inspect the slings and shackles in the rigging register.
4. **Well control.** Record the slow-rate pressures, then prepare the kill sheet so the numbers are ready in advance.
5. **Reporting.** Log a near miss from the day with its risk level and corrective actions, and watch it appear in the Safety Dashboard.
6. **Month end.** Fill in the HSE Monthly Report, with the safety rates worked out for you, and export it.

## 6. Engineering Highlights

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The parts of the project that took the most thought, and the reasoning behind each.

| Challenge                                | Approach                                                                                                                                  | Why it matters                                                                 |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Dependent calculations                   | Eight (or nine) calculations always run in dependency order and re-run when a value changes or a saved sheet is opened                    | Every formula has its inputs ready; a loaded sheet is always consistent        |
| Rounding that errs safe                  | Strokes, times and kill mud weight round up; maximum mud weight and MAASP round down                                                      | A limit is never overstated                                                    |
| A PDF that matches a paper form          | A dense two-page A4 layout reproduces the IWCF sheet, including rotated labels and an embedded chart                                      | Users can print a familiar document                                            |
| One save pattern everywhere              | Numbered slots with a label and time kept apart from the bulky form contents                                                              | The slot picker opens instantly, and a new feature only needs a new table name |
| Growing a database safely                | Seven versions, each only adding tables, created only if missing, with a one-time migration of older file saves                           | No data loss on upgrade                                                        |
| Six permits, one form                    | A shared five-section structure with each permit supplying only its own activity lists, hazards and checklist groups                      | Adding a new permit is mostly a matter of new lists                            |
| Bundling unlike records                  | The composite export reads saved permits and crane plans from different tables and builds each page like the original                     | A whole job's paperwork in one PDF                                             |
| Judging risk consistently                | Severity $\times$ likelihood gives a score and band; automatic High Potential rules can be overridden; target dates follow the risk       | Reports are comparable and reviews are prompted                                |
| Three record types, one dashboard        | Observations, near misses and incidents are converted to one common shape                                                                 | One table and one set of filters cover everything                              |
| A failed inspection can't be argued away | A critical defect or any failed minor item forces an overall FAIL that cannot be overridden                                               | The record reflects the worst finding                                          |
| An editable SIMOPS matrix                | Twenty-five operations against each other with Not permitted, Permitted and Permitted with approval, plus undo, redo and rotated headings | Practical to adapt on a phone                                                  |
| Graphics without a chart library         | The RPM and flow graphs, the well diagram, the sling-angle diagram and the crane diagrams are drawn directly                              | Full control of the look                                                       |
| One page that fits any screen            | The Delegation of Authority letter is built at a fixed width and scaled uniformly                                                         | The layout never breaks                                                        |

## 7. Design Decisions

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- **Offline-first.** A rig can be far from a signal, so every tool works on the device.
- **Colour with meaning.** Cyan for data, red for critical work, amber for drill-floor operations, green as the main accent and grey for “coming soon”.
- **A dark theme.** High contrast suits use in poor light and with gloves.
- **One screen pattern.** A header, a breadcrumb line, numbered collapsible sections and a shared bottom bar.
- **Honest about unfinished work.** Planned modules appear as “coming soon” tiles, so the shape of the finished product is visible.
- **Reference and calculation together.** Alongside the calculators sit visual references, such as the lifting manual and the mast inspection guide.

## 8. What This Project Demonstrates

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- **Domain knowledge** — well control, MWD, permits to work, safety reporting and lifting operations, turned into working software.
- **Product thinking** — a clear user, a consistent pattern and visible planning of what comes next.
- **Calculation engineering** — dependent formulas, unit handling and safe rounding.
- **Data modelling and persistence** — one database, a repeatable save pattern, safe upgrades and a migration.
- **Document generation** — PDFs that mirror real forms, plus Excel and CSV exports.
- **Interface craft** — custom-drawn graphics, inline pickers, editable grids with undo and redo.
- **Scale and consistency** — about 181,000 lines across 46 files that still behave like one app.
- **Documentation** — 27 guides that explain each part without exposing the code.

## 9. Where It Could Go Next

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Ideas for future versions, not commitments:

- Build out the “coming soon” modules: the shut-in simulators, kick detection, the well-control methods, the Risk Assessment and HAZOP reports, and the Daily and Weekly HSE reports.
- Link the Radiography permit into the permit hub.
- Add LWD, directional drilling and further operations departments.
- Complete the lug calculation in the rigging module.
- Move the remaining file-based saves into the database so every module shares one storage pattern.
- Optional encrypted backup and restore of saved sheets.

## 10. Companion Guides

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Each part of the app has its own guide:

| Guides                                                                                              | Covers                   |
|-----------------------------------------------------------------------------------------------------|--------------------------|
| App Overview · Setup & Startup · Home & Navigation · Data Storage                                   | Foundations              |
| MWD Survey · MWD Downlinking · Rig Drill Sensors                                                    | The training simulations |
| Well Control Hub · SCR Measure (surface, subsea) · Kill Sheet (surface, subsea)                     | Well control             |
| Safety Hub & Dashboard · Report An Event · Incident · Near Miss · Observations · HSE Monthly Report | Safety reporting         |
| Checklists Part 1 and Part 2                                                                        | The seven checklists     |
| Permit to Work and its three permit guides                                                          | Permits to work          |
| Crane Lifting Plan · Lifting Manual · Rigging & Slings                                              | Lifting operations       |